

No. of copies = 80

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PYMBLE LADIES' COLLEGE

YEAR 12

TRIAL HIGHER SCHOOL CERTIFICATE - 1995

3/4 UNIT MATHEMATICS

Time Allowed: 2 Hours

Plus 5 Minutes reading time

DIRECTIONS TO CANDIDATES:

- * Attempt all questions.
- * All questions are of equal value.
- * All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- * Each question is to be started on a new page and labelled with your name and your teacher's name.
- * Standard integrals are attached.
- * Approved calculators may be used.

PLC 1985

Question 1

Marks

- (a) If $f(x) = x \sin^{-1} x$ evaluate $f'(\frac{\sqrt{3}}{2})$ 3
- (b) (i) Two dice are tossed and the two numbers are added. State the probability that this total is 10 or greater. 3
- (ii) Two dice are tossed eight times. Find the probability that a total of 10 or greater occurs:
- (α) at least once
- (β) exactly once
- (give answers correct to 3 significant figures)
- (c) (i) Use the factor theorem to factorise $p^3 + 2p + 12$ 6
- (ii) $P(2p, p^2)$ is a point on the parabola $x^2 = 4y$
- (α) show that the equation of the normal at P is $x + py = p^3 + 2p$
- (β) Find the co-ordinates of P if this normal cuts the x axis at $(-12, 0)$

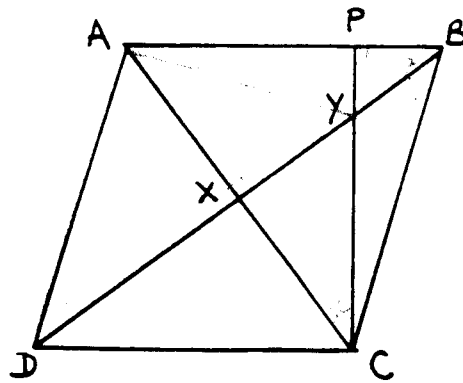
Question 2 (Start a new page)

- (a) Find $\int \frac{x dx}{1 + 2x}$ using the substitution $u = 1 + 2x$ 4
- (b) (i) Evaluate $\int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta$ 8
- (ii) Use the substitution $x = \cos 2\theta$ to evaluate $\int_{\frac{1}{2}}^1 \sqrt{\frac{1-x}{1+x}} dx$

Question 3 (Start a new page)

Marks

(a)



5

ABCD is a rhombus whose diagonals intersect at X. The perpendicular CP from C to AB cuts BD at Y.

Prove that:

- (i) points B, P, X, C are concyclic
- (ii) points A, Y, C, D are concyclic

(b) (i) Sketch $y = 4 \tan^{-1}x$

7

- (ii) By drawing a suitable line, l , on this graph, show that the equation

$$4 \tan^{-1}x + x - 4 = 0$$

has a root close to $x = 1$.

Clearly state the equation of l .

- (iii) Use Newton's method once to find a better approximation to this root. (Give your answer correct to 3 decimal places).

Question 4 (Start a new page)

- (a) A population of bacteria is treated with a new drug. B , the number of bacteria present after t minutes is given by

6

$$B = 200 + 500 e^{kt}$$

If initially the population decreases at the rate of 10 bacteria/minute

- (i) evaluate k , and
 - (ii) find the time it takes for the population to halve.
 - (iii) Sketch the graph of B against t .
- (b) A function $f(x)$ is defined as
- $$f(x) = x^3 + 6x^2 + 12x - 14 \text{ for } -1 \leq x \leq 4$$
- (i) Show that $f(x)$ is an increasing function.
- (ii) Explain why the inverse function $y = f^{-1}(x)$ exists and state its domain and range.
- (iii) Find the gradient of $y = f^{-1}(x)$ at the point $(5, 1)$ on it.

6

Question 5 (Start a new page)

Marks

(a) Given $x(1+x)^n = \sum_{r=0}^n \binom{n}{r} x^{r+1}$

3

show by differentiation that

$$\binom{n}{0} + 2 \binom{n}{1} + 3 \binom{n}{2} + \dots + (n+1) \binom{n}{n} = (n+2) 2^{n-1}$$

(b) A particle is oscillating in simple harmonic motion on a straight line. Its displacement, x cm, from the origin O at time t seconds

9

is given by $x = 3 + 2 \cos\left(\frac{1}{2}t + \frac{\pi}{3}\right)$

- (i) Find its initial position.
- (ii) State the frequency and the amplitude of the oscillation.
- (iii) Find the first two times that the particle moves through its centre of oscillation.
- (iv) If its velocity at time t seconds is v cm/sec, express v^2 as a function of x .

Question 6 (Start a new page)

(a) $A(t, e^t)$ and $B(-t, e^{-t})$ are two points on $y = e^x$, ($t > 0$). The tangents at A and B intersect at an angle of 45°

6

- (i) Show that $e^t - e^{-t} = 2$
- (ii) Solve this equation for t

(b) When x cm from the origin, the acceleration of a particle moving on a straight line is given by

6

$$\frac{d^2x}{dt^2} = \frac{-4}{(x+1)^3}$$

It has an initial velocity of 2 cm/second at $x = 0$

- (i) If its velocity is V cm/second, show that $V = \frac{2}{x+1}$
- (ii) Find the time taken for the particle to reach $x = 4$

Question 7 (Start a new page)

Marks

- (a) Solve $\sin \alpha - 3 \cos \alpha = 3$ for $0^\circ \leq \alpha \leq 360^\circ$

4

(b)

8

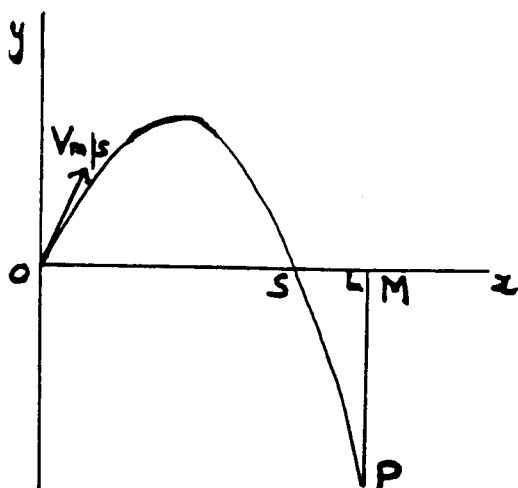


Diagram not to scale

A projectile is fired from a point O with initial speed of V m/s at an angle of elevation θ . If x and y are the horizontal and vertical displacements of the projectile in metres from O at time t seconds

later then $x = Vt \cos \theta$ and $y = Vt \sin \theta - \frac{1}{2}gt^2$

where g m/s² is the acceleration due to gravity.

The projectile falls to a point P below the level of O such that $PM = OM$.

- (i) Prove that the time taken to reach P is $2V \frac{(\sin \theta + \cos \theta)}{g}$ seconds

- (ii) Show that the distance OM is

$$\frac{V^2}{g} (\sin 2\theta + \cos 2\theta + 1) \text{ metres}$$

- (iii) If the horizontal range of the projectile level with O (i.e. OS) is r metres and $OM = \frac{4r}{3}$, ($r > 0$)

Prove that $\sin 2\theta - 3 \cos 2\theta = 3$

- (iv) Hence by using part (a), find the value of θ .

- (v) If the magnitude of the velocity of the projectile at P is KV m/s find the exact value of K .

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

1 (R) $f(x) = x \arcsin^{-1} x$

(3) $f'(x) = \arcsin^{-1} x + \frac{x}{\sqrt{1-x^2}}$

$$f'\left(\frac{\sqrt{3}}{2}\right) = \arcsin^{-1} \frac{\sqrt{3}}{2} + \frac{\frac{\sqrt{3}}{2}}{\sqrt{1-\frac{3}{4}}}$$

$$= \frac{\pi}{3} + \sqrt{3}$$

(b) (i) $\frac{1}{6}$

(3) (ii) (a) $P(\text{at least once}) = 1 - \left(\frac{5}{6}\right)^8$
 $= 0.767$

(b) $P(\text{exactly once}) = {}^8C_1 \left(\frac{1}{6}\right) \left(\frac{5}{6}\right)^7$
 $= 0.372$

(c) (i) $f(p) = p^3 + 2p + 12$
 $f(-2) = -8 - 4 + 12 = 0$
 $\therefore (p+2)$ is a factor

$$f(p) = (p+2)(p^2 - 2p + 6)$$

(ii) $x = 2p$ $y = p^2$
 $\frac{dx}{dp} = 2$ $\frac{dy}{dp} = 2p$

$$\therefore \frac{dy}{dx} = p$$

grad. of normal $= -\frac{1}{p}$

$$\begin{aligned} \text{Eqn 9 normal: } y - p^2 &= -\frac{1}{p}(x - 2p) \\ py - p^3 &= -x + 2p \\ x + py &= p^3 + 2p \end{aligned}$$

$(-12, 0)$ lies on normal

$$\begin{aligned} -12 &= p^3 + 2p \\ p^3 + 2p + 12 &= 0 \end{aligned}$$

$\therefore p = -2$ is only soln.

$$P(-4, 4)$$

2. a) $\int \frac{x dx}{1+2x}$

$u = 1+2x$
 $du = 2dx$

$= \int \frac{u-1}{4u} du$

$= \frac{1}{4} \int (1 - \frac{1}{u}) du$

$= \frac{1}{4} (u - \ln u) + C$

$= \frac{1}{4} (1+2x) - \frac{1}{4} \ln(1+2x) + C$

(b) i) $\int_0^{\pi/6} \sin^2 \theta d\theta$
 $= \frac{1}{2} \int_0^{\pi/6} (1 - \cos 2\theta) d\theta$
 $= \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/6}$
 $= \frac{\pi}{12} - \frac{\sqrt{3}}{8}$

(ii) $\int_{1/2}^1 \sqrt{\frac{1-x}{1+x}} dx$

$x = \cos 2\theta$
 $dx = -2 \sin 2\theta d\theta$
 $x=1 \quad \theta=0$
 $x=1/2 \quad \theta=\pi/6$

$= \int_{\pi/6}^0 \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} \cdot -2 \sin 2\theta d\theta$

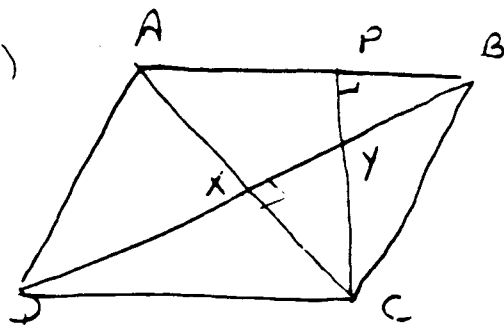
$= \int_{\pi/6}^0 \sqrt{\frac{2 \sin^2 \theta}{2 \cos^2 \theta}} \cdot -2 \sin 2\theta d\theta$

$= 4 \int_{\pi/6}^0 \frac{\sin \theta}{\cos \theta} \times \sin \theta \cos \theta d\theta$

$= 4 \int_{\pi/6}^0 \sin^2 \theta d\theta$

$= \frac{\pi}{3} - \frac{\sqrt{3}}{2}$

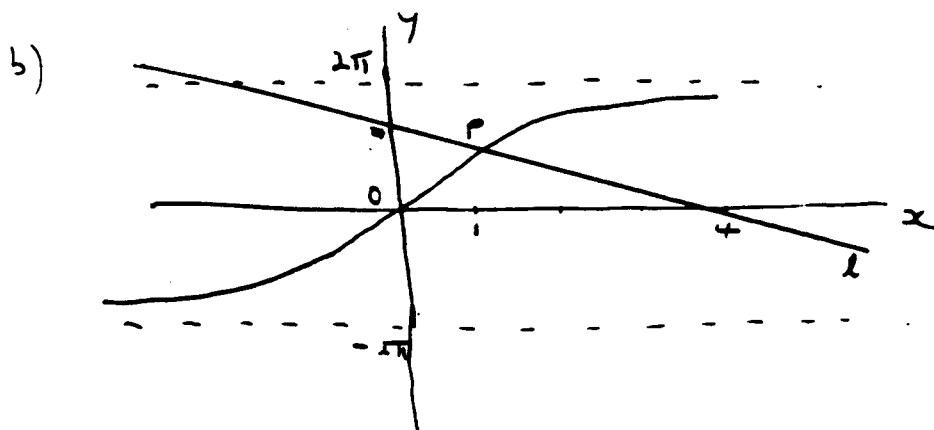
3 (a)
(5)



Proof (i) $\angle BXC = 90^\circ$ (diags of rhombus are $\perp$)
 $\angle BPC = 90^\circ$ (given)
 $\therefore$ pts B, P, X, C are concyclic (equal $\angle$'s on BC & on same side of BC)

(ii) $AB = AD$ (sides of rhombus)
 $\therefore \angle ABD = \angle ADB$ (base $\angle$'s of isosceles $\triangle ABD$)
 $\angle PBX = \angle PCX$ (angles in same segment of circle BPCX)
 $\therefore \angle ADY = \angle ACY$

$\therefore$ pts A, Y, C, D are concyclic (equal $\angle$'s on AY & on same side of AY)



1, $y = 4 \tan^{-1} x$
 + asymptote

1, line l + scale

l is line $y = 4 - x$

P, point of intersection, has x value close to

$$f(x) = 4 \tan^{-1} x + x - 4$$

$$f'(x) = \frac{4}{1+x^2} + 1$$

$$f(1) = 4 \tan^{-1} 1 - 3$$

$$= \pi - 3$$

$$f'(1) = 3$$

$$x_1 = 1$$

$$x_2 = 1 - \frac{\pi - 3}{3}$$

$$= 0.953 \text{ (3dp)}$$

4. a) $B = 200 + 500 e^{kt}$

⑥ $t=0 \quad \frac{dB}{dt} = -10$

$\frac{dB}{dt} = 500 k e^{kt}$

$\therefore -10 = 500 k$

$k = -\frac{1}{50}$

①

n) Initially $B = 700$
When $B = 350$

$350 = 200 + 500 e^{-\frac{t}{50}}$

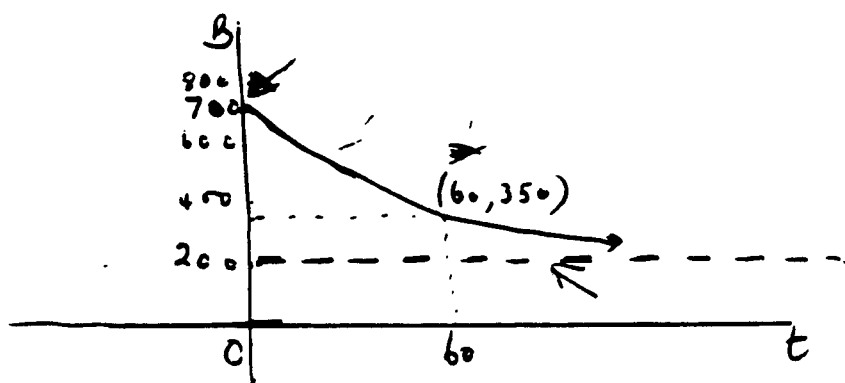
$e^{-\frac{t}{50}} = \frac{150}{500}$

①

$t = 60.2$

It takes 60.2 minutes.

①



$t \rightarrow 0$
scale
disproportion

(b) $f(x) = x^3 + 6x^2 + 12x - 14$, $-1 \leq x \leq 4$

(6) $f'(x) = 3x^2 + 12x + 12$
 $= 3(x+2)^2$

Since $(x+2)^2 > 0$ for $-1 \leq x \leq 4$ ∴

$f'(x)$ is positive

∴ $f(x)$ is an increasing fn

(i) Since $f(x)$ is an increasing fn
 it is a 1-1 fn
 ∴ it has an inverse

$f^{-1}(x)$ $D: -21 \leq x \leq 194$
 $R: -1 \leq y \leq 4$

(ii) $f'(5) = 27$

grad of inverse fn at $x = 5$ is $\frac{1}{27}$

$$5(a) \quad x(1+x)^n = \sum_{r=0}^n \binom{n}{r} x^{r+1}$$

(3)

$$1 = \binom{n}{0}x + \binom{n}{1}x^2 + \binom{n}{2}x^3 + \dots + \binom{n}{n}x^{n+1}$$

Differentiate

$$1 \cdot (1+x)^n + nx(1+x)^{n-1} = \binom{n}{0} + 2\binom{n}{1}x + 3x^2\binom{n}{2} + \dots + (n+1)x^n$$

Let $x=1$

$$1 \cdot 2^n + n \cdot 2^{n-1} = \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \dots + (n+1)\binom{n}{n}$$

$$2^{n-1}(2+n) = \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \dots + (n+1)\binom{n}{n}$$

(b) $x = 3 + 2 \cos\left(\frac{1}{2}t + \frac{\pi}{3}\right)$

(9)

(i) Initially $t=0$

$$x = 3 + 2 \cos \frac{\pi}{3} = 4$$

Particle is 4 cm to right of O

(ii) Frequency = $\frac{1}{4\pi}$ osc/sec

Amplitude = 2 cm.

(iii) Centre of oscillation $x=3$

$$3 + 2 \cos\left(\frac{1}{2}t + \frac{\pi}{3}\right) = 3$$

$$\textcircled{1} \quad 2 \cos\left(\frac{1}{2}t + \frac{\pi}{3}\right) = 0$$

$$\textcircled{2} \quad t = \frac{\pi}{3} \text{ and } \frac{7\pi}{3}$$

(iv) $v = -\sin\left(\frac{1}{2}t + \frac{\pi}{3}\right)$

$$v^2 = \sin^2\left(\frac{1}{2}t + \frac{\pi}{3}\right)$$

$$= 1 - \cos^2\left(\frac{1}{2}t + \frac{\pi}{3}\right)$$

$$= 1 - \frac{(x-3)^2}{4}$$

6 (a)

$$y = e^x$$

$$\frac{dy}{dx} = e^x$$

at A, grad = $e^t = m_1$

at B, grad = $e^{-t} = m_2$

$$\tan \theta = 1$$

θ is angle between tangents

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$1 = \frac{e^t - e^{-t}}{1 + 1}$$

2

$$\therefore 2 = e^t - e^{-t}$$

$$e^t - \frac{1}{e^t} = 2$$

1

$$e^{2t} - 2e^t - 1 = 0$$

1

$$e^t = \frac{2 \pm \sqrt{4+4}}{2}$$

$$= 1 \pm \sqrt{2}$$

1

$$e^t = 1 + \sqrt{2} \quad \text{since } e^t > 0$$

1

$$\therefore t = \ln(1 + \sqrt{2})$$